

Chapter 32 — Embryology of the Human Genitourinary Tract — Numbers, Gold Standards and Traps

Source pp. 584–607 only (partial coverage): the rest of bladder/ureter, kidney and closing sections were not supplied

IF YOU READ NOTHING ELSE

- **SRY (sex-determining region Y), short arm of Y** → Sertoli cells → testis; no SRY → ovary by default
- **Sertoli** → **Müllerian-inhibiting substance (MIS)** (Müllerian regression, weeks 8–10); **Leydig** → **testosterone** from 8 weeks
- **Type 2 5 α -reductase** (testosterone → dihydrotestosterone, DHT): needed for penis + prostate; deficiency → severe hypospadias, rudimentary prostate
- **Hypospadias** $\approx$ 1 in 250 boys; commonest meatus at **shaft-glans junction**; developmental arrest
- **Only fold fusion is androgen-dependent**; plate canalisation is androgen-independent
- **Plication at 12 o'clock**: nerve-free zone + thickest tunica (dorsal nerves 11 and 1 o'clock)
- **Cryptorchidism** $\approx$ 3% newborn males; hydrocele / hernia more common
- **Streak gonad 46,XY** → **remove** (gonadoblastoma); all streaks → hormone replacement
- **Entire vaginal epithelium + hymen = urogenital sinus epithelium (UGE)** (Bulmer); imperforate hymen = commonest vaginal obstruction
- **Weigert-Meyer**: upper-pole orifice caudal + medial → ectopia; lower-pole lateral → reflux

EXTERNAL GENITALIA

- **Indifferent stage** 8–9 weeks; development mostly complete by end of 2nd trimester
- **First sign of masculinisation**: $\uparrow$ anogenital distance → tubercle elongation → urethral plate
- **Germ layers**: ectoderm = skin / prepuce; mesoderm = corpora; endoderm = urethral plate
- **Whole urethral epithelium = endoderm (urogenital sinus, UGS)**; meatus = ectoderm–endoderm junction
- **Double zipper**: canalisation (proliferation, no apoptosis) → fold fusion (median raphe)
- **Timetable**: coronal sulcus 11–12 wk; folds fused 16 wk; penis complete 18 wk; fetal chordee gone 20 wk
- **Shaft urethra** = fusion; **glanular urethra** = direct canalisation
- **Glenister** = outdated ectodermal-intrusion theory of glanular urethra
- **Hypospadias = 4 components**: ectopic meatus, dorsal hooded foreskin, ventral skin deficiency, $\pm$ curvature
- **Hypospadias aetiology**: mostly unknown; genetic susceptibility + endocrine disruptors
- **Hypospadiac penis normal** except urethra, spongiosum, ventral skin, ventral prepuce
- **Hooded foreskin**: prepuce absent ventrally, excess dorsally
- **Leydig cells + testosterone**: 8 weeks
- **Nerves**: somatic S2–S4 pudendal via **Alcock's canal**; sympathetic L1–L2; parasympathetic S2–S4
- **Neural crest** → peripheral / autonomic nerves at week 5
- **Cavernous nerves** exit between transverse perineal muscles and membranous urethra
- **At 16 weeks**: penis $\approx$ 90° to body; clitoris recessed to body wall
- **Duplicated urethra**: plate splits (complete from bladder; Y along shaft); Sonic hedgehog
- **Clitoris = penis without urethra**; vestibular folds never fuse; 12 o'clock nerve-free zone too
- **Congenital adrenal hyperplasia (CAH)**: autosomal recessive, 46,XX virilised; urethra may be male-identical
- **CAH clues**: impalpable gonads + hyperpigmentation; mimics hypospadias + bilateral impalpable testes

PROSTATE AND SEMINAL VESICLE

- **Prostatic buds 10 weeks** (9.5–10) from UGE below bladder; branching from $\sim$ 11 weeks
- **Buds mainly lateral to verumontanum**; budding stochastic — fetal ventral buds lost
- **Six stages**: prebud → solid buds → branching → canalisation → luminal / basal → secretion
- **FOXA1** = endodermal (UGE) marker
- **Verumontanum** = Wolffian ducts + utricle (mesoderm) meet UGE (endoderm); uniquely human
- **Canalisation** proximal → distal; **smooth muscle** ventral first, then lateral / dorsal
- **DHT** $\approx$ 10 $\times$ androgen receptor (AR) affinity of testosterone; 5 α R2 mainly in mesenchyme
- **Non-functional AR** → **no prostate**; female UGS + androgen → prostate
- **5 α R1 normal** in 5 α R2 deficiency — does not compensate
- **Jost and Price**: animal studies establishing androgen dependence
- **Ductal length** to first branch $\approx$ 250 μ m (14 wk) → 1000 μ m (17 wk)
- **Urogenital sinus mesenchyme (UGM) instructs epithelium** — induces prostate even in bladder epithelium
- **Table 32.1**: basal = TP63, KRT5/14, KRT19, GSTpi; luminal = KRT8/18
- **Oestrogen (DES)** → **squamous metaplasia** via ESR1 (oestrogen receptor α), needed in stroma *and* epithelium
- **Seminal vesicle** = Wolffian diverticulum 10–12 wk; side ducts months 4–5; innervated 13 wk; main anomaly = cyst

GONADS AND GONADAL DESCENT

- **Primordial germ cells (PGCs)**: yolk sac week 3 → hindgut dorsal mesentery → ridge; germinal ridge 4 wk
- **Primary sex cords** week 6; testicular cords $\sim$ 6 wk; ovarian cortex / medulla $\sim$ 8 wk
- **Gonads look identical** 7–8 wk; then testis spherical, ovary oblong
- **Ovary = default**: no SRY → no Sertoli cells, MIS, Leydig cells or androgen
- **Seminiferous cords** appear to come from mesonephric tubules, not primary sex cords

- **Efferent ductules** = 5–12 mesonephric tubules; full tubule-epididymis continuity by 3rd month
- **Testosterone control:** placental chorionic gonadotrophin early → fetal pituitary later
- **Wolffian differentiation** weeks 8–12 → epididymis, vas, seminal vesicle
- **Remnants:** appendix testis + utricles = Müllerian; appendix epididymis + paradidymis = Wolffian / mesonephric
- **Germ cell nests** 10–11 wk (2–4 cells); primary follicle ~20 wk; secondary at term; antral + theca postnatal
- **Oocytes arrest in meiosis I** from fetal life to puberty
- **Homologues:** Cowper ↔ Bartholin; Littre ↔ Skene; spongiosum ↔ vestibular bulb; **Gartner duct** = female Wolffian remnant
- **Transabdominal phase** weeks 10–15: cranial suspensory ligament (CSL) regresses (androgen); gubernacular swelling (androgen-independent; INSL3)
- **Inguinoscrotal phase** from 7th month: androgen via genitofemoral nerve CGRP (calcitonin gene-related peptide); migration 3–5 cm
- **INSL3** (insulin-like peptide 3, 1993): null mice = isolated intra-abdominal cryptorchidism
- **Operate early:** early orchidopexy restores fertility potential (Insl3-null mice)
- **Gubernacular cord transection** → contralateral canal or aberrant intra-abdominal testis
- **Both gonads in pelvis** by month 3; testis in scrotum ~month 7; ovary on broad ligament month 3
- **Female gubernaculum** → ovarian ligament + round ligament; patent processus → indirect hernia
- **Anomaly map:** CSL retained → intra-abdominal; failed migration → congenital cryptorchidism; failed closure → hydrocele / hernia; failed involution → ascending testis
- **Streak gonad:** 45,X (Turner) / 46,XX / 46,XY; Müllerian retained, Wolffian regressed
- **Ovotesticular:** ducts follow ipsilateral gonad; female fertility possible, no spermatogenesis reported
- **Persistent Müllerian duct syndrome (PMDS):** 46,XY + uterus / tubes ("hernia uteri inguinale"); usually MIS receptor defect; found at exploration for bilateral abdominal testes
- **Müllerian sensitivity window** to MIS 9–13 weeks

MÜLLERIAN STRUCTURES

- **Müllerian invagination** start week 5; **Wolffian duct** = essential guidewire
- **Ducts between Wolffian ducts** weeks 7–8; **fuse week 8** (uterovaginal canal); **septum gone week 9**
- **Indifferent stage** ≈ 54 days (Carnegie stage 22)
- **Uterine tube** = cranial unfused Müllerian ducts; fimbriae from ostia
- **Endocervix** simple columnar; **exocervix** stratified squamous
- **Markers:** PAX2 = Müllerian; FOXA1 = UGE; switch abruptly at vaginal-cervical junction
- **Vaginal canalisation** ~15 wk → mostly complete 18–19 wk; glycogenated by 21 wk
- **Koff** = mixed, Müllerian-predominant; **Bulmer** = all UGE → supported by immunohistochemistry

- **Mouse is opposite:** Müllerian epithelium replaces sinus vagina
- **Transverse vaginal septum** ≈ 1 in 80,000; middle / upper third junction (failed canalisation)
- **Didelphys** = failed fusion; **unicornuate** = one duct absent; **septum persists** → obstruction / duplication
- **Obstructed hemi-vagina and ipsilateral renal anomaly (OHVIRA):** post-menarche cyclic pain + discharge; may present as acute abdomen / tubo-ovarian abscess
- **OHVIRA kidney** usually dysplastic / atrophic ± ectopic ureter; true agenesis = absent hemi-trigone

BLADDER: URETERAL DEVELOPMENT

- **Urorectal septum divides cloaca** weeks 5–6 → dorsal anorectal canal + ventral UGS (Rathke, Tourneux)
- **Allantois** → urachal remnant
- **Anorectal malformations** from cloacal membrane, not urorectal septum
- **Ureterovesical junction** inserts into trigone identically in both sexes
- **Wolffian entry to cloaca** = landmark: vesicourethral canal (bladder, pelvic urethra) vs caudal UGS (phallic urethra; female urethra + vestibule)
- **Common excretory ducts join UGS** weeks 5–6; apoptosis separates ureter from mesonephric duct
- **Trigone** mostly bladder smooth muscle; orifices migrate cranially + laterally
- **Bud too low** → lateral orifice → reflux; **too high** → caudal / medial → ectopic
- **Ectopic sites:** male — bladder neck, verumontanum, vas; female — Gartner duct cyst, vestibule
- **Prostate below ureteric insertion** → larger male prostatic urethra

CLASSIC TRAPS

- **Trap:** glanular urethra ectodermal — **Actually:** whole urethra endodermal (UGS)
- **Trap:** hypospadias = deformation — **Actually:** developmental arrest
- **Trap:** MIS from Leydig cells — **Actually:** Sertoli; Leydig make testosterone
- **Trap:** appendix testis Wolffian — **Actually:** Müllerian; appendix epididymis is Wolffian
- **Trap:** gubernacular swelling needs androgen — **Actually:** androgen-independent; CSL regression is dependent
- **Trap:** ascending testis congenital — **Actually:** acquired; processus fails to involute
- **Trap:** PMDS = absent MIS — **Actually:** usually MIS receptor defect
- **Trap:** OHVIRA = renal agenesis — **Actually:** renal anomaly, usually dysplasia; check for ectopic ureter before septum resection
- **Trap:** adult upper vagina Müllerian — **Actually:** UGE replaces it up to cervix
- **Trap:** fetal prostatic squamous metaplasia pathological — **Actually:** normal, maternal oestrogen